Project: Worked examples
Title: Motor start voltage dip and PF correction
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1. Scope and method

The voltage dip when a three-phase induction motor starts, at the motor terminals and at the point of common coupling, from the supply impedance (a fault level or a transformer), the cable and the motor's locked-rotor impedance for a direct-on-line or star-delta start (a passive motor impedance) or a soft-start or drive start (a regulated current), by the complex potential divider; the dip is checked against a user limit (default 10 %). Also the capacitor kvar to raise the running power factor to a target, the capacitor current and the parallel-resonance order, with a warning when harmonic loads are present. Steady-state phasor model: no motor run-up dynamics, no flicker assessment, no starting-torque adequacy check.
Rated current: $I_n = P/(\sqrt{3}\cdot U\cdot\cos\varphi\cdot\eta)$, or entered Starting current at rated voltage: DOL $I_{LR}\cdot I_n$; star-delta $I_{LR}\cdot I_n/3$; soft starter $\min(k_{soft}\cdot I_n, I_{LR}\cdot I_n)$; drive $k_{vfd}\cdot I_n$ Motor starting impedance (DOL, star-delta, and a soft starter that does not reach its limit): $|Z_m| = (U/\sqrt{3})/I_{start}$, for a soft starter $(U/\sqrt{3})/(I_{LR}\cdot I_n)$, at the starting power factor, $Z_m = |Z_m|\cdot(\cos\varphi_{st} + j\sin\varphi_{st})$ Regulated current (a drive; a soft starter when its fully conducting current $E/|Z_s + Z_c + Z_{LR}|$ exceeds $k_{soft}\cdot I_n$): $|I| = I_{start}$ held, lagging V_m by φ; with $R + jX = Z_s + Z_c$, $a = |I|(R\cos\varphi + X\sin\varphi)$, $b = |I|(X\cos\varphi - R\sin\varphi)$: $|V_m| = \sqrt{(E^2 - b^2)} - a$, $V_{pcc} = V_m + I\cdot Z_c$, and $Z_m = V_m/I$ is the equivalent impedance Source impedance: transformer $|Z_T| = u_k\cdot U^2/S_n$, plus an optional upstream network $|Z_Q| = U^2/S_{kQ}$ in series; fault level $|Z_s| = c\cdot U^2/S_k$ or $c\cdot U/(\sqrt{3}\cdot I_k)$; each element $Z = |Z|\cdot(1 + j\cdot X/R)/\sqrt{(1 + (X/R)^2)}$ Cable: $R = \rho_T\cdot L/(A\cdot n)$ with $\rho_T = \rho_{20}\cdot[1 + \alpha(T_c - 20)]$, or $R'\cdot L/n$; $X = X'\cdot L/n$ (as E3) Phasor divider for a passive motor (phase voltage $E = U/\sqrt{3}$ at 0°): $I = E/(Z_s + Z_c + Z_m)$; $V_{pcc} = E\cdot(Z_c + Z_m)/(Z_s + Z_c + Z_m)$; $V_m = E\cdot Z_m/(Z_s + Z_c + Z_m)$ Dip: $100\cdot(1 - |V|/E)$ at the motor terminals and at the point of common coupling; checked against the limits Cable drop along V_m: $dV_{cab} = \sqrt{3}\cdot\mathrm{Re}[I\cdot Z_c\cdot\bar{V}_m/|V_m|] = \sqrt{3}\cdot I\cdot(R\cos\varphi_{st} + X\sin\varphi_{st})$, the E3 three-phase form at the dipped current Power-factor correction: $P_{in} = P/\eta$ (or $\sqrt{3}\cdot U\cdot I_n\cdot\cos\varphi$ when I_n is entered), $Q_c = P_{in}\cdot(\tan\varphi_1 - \tan\varphi_2)$ with φ_1 the motor's (a drive's displacement factor with a drive), $I_c = Q_c/(\sqrt{3}\cdot U)$, $I_{after} = P_{in}/(\sqrt{3}\cdot U\cdot\cos\varphi_2)$ Resonance: $S_{sc} = U^2/|Z_s|$, parallel resonance order $h \approx \sqrt{(S_{sc}/Q_c)}$ ($|Z_s|$ used for X_s, so h reads 1 to 1.4 % low)

Ref: IEC 60909-0 cl. 5.3 — Voltage factor c and the equivalent voltage source
Ref: IEC 61000-3-11 scope — Limitation of voltage changes and flicker in public low-voltage systems (limits are user inputs)
Ref: Hughes, Electrical and Electronic Technology Ch. on AC circuits — Complex impedance and the potential divider
Ref: Hughes, Electrical and Electronic Technology Ch. on power factor — Power-factor correction: $Q = P(\tan\varphi_1 - \tan\varphi_2)$

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
U	Supply and motor rated voltage (line-to-line)	400	V	spec: Site supply SS-1: 400 V three-phase	50 … 1000 V
source	Source impedance from	Transformer		drawing: E-SCH-031 rev A	any
fault_basis	Fault level given as	Fault level S_k (MVA)		default	any
S_k	Three-phase fault level at the point of supply (use the minimum fault level, which gives the deepest dip)	10	MVA	default	0.5 … 500 MVA

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I_k	Three-phase fault current at the point of supply (links E5 I_k3)	15	kA	default	1 … 150 kA
S_n	Transformer rating	500	kVA	spec: Transformer datasheet TR-01: 500 kVA rating plate	25 … 5000 kVA
u_k	Transformer impedance voltage	4	%	reference: record: transformer TR-01 impedance voltage 4 % (rating plate; test value invented for this example)	1 … 12 %
S_kQ	Upstream network fault level at the transformer primary, in series with the transformer (minimum fault level; blank: infinite bus)	15	MVA	spec: Network operator letter NO-7: minimum fault level 15 MVA at the transformer primary (test value invented for this example)	1 … 5000 MVA
XR_Q	Upstream network X/R ratio	10		assumption: Network X/R taken as 10, mostly reactive.	1 … 50
XR	Source X/R ratio (of the transformer, or of the fault level)	4.2		reference: record: transformer TR-01 X/R 4.2 (from the load-loss figure; test value invented for this example)	0.5 … 40
P	Motor rated output (shaft) power, from the nameplate (E1 P_cont is only a lower bound)	160	kW	spec: Pump schedule PS-05: 160 kW motor rated output (test value invented for this example)	0.1 … 2000 kW
eta	Motor efficiency at rated load	0.95		spec: Motor datasheet MD-21: 95 % at full load (test value invented for this example)	0.5 … 0.99
cos_phi	Motor power factor at rated load	0.88		spec: Motor datasheet MD-21: 0.88 at full load (test value invented for this example)	0.5 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
I_LR_r	Locked-rotor current ratio I_LR/I_n	7		reference: record: motor MD-21 locked-rotor current ratio 7.0 (datasheet; test value invented for this example)	3 … 10
cos_phi_st	Power factor at locked rotor (starting)	0.3		reference: record: motor MD-21 locked-rotor power factor 0.3 (datasheet; test value invented for this example)	0.1 … 0.8
starter	Starter type	Star-delta		drawing: E-SCH-031 rev A	any
k_soft	Soft-starter current limit as a multiple of the rated current	3.5		default	1 … 6
k_vfd	Drive line-current limit as a multiple of the rated current	1.5		default	0.5 … 2.5
cos_phi_vfd	Drive line displacement power factor while starting	0.95		default	0.5 … 1
cable_data	Cable resistance from	resistivity and area		spec: Cable schedule CS-31: resistivity route as E3	any
L_cab	Cable length, source to motor (links E4 L)	60	m	drawing: E-LAY-031 rev B	0 … 1000 m
n_par	Parallel conductors per phase	1		drawing: E-SCH-031 rev A	1 … 6
A_c	Conductor cross-section (links E4 CSA)	95	mm^2	spec: Cable schedule CS-31: 95 mm² copper	1 … 1000 mm^2
rho_20	Resistivity at 20 °C (copper 0.01724, aluminium 0.0282 Ω·mm²/m)	0.01724	ohm*mm^2/m	reference: record: annealed copper 1.724e-8 Ω·m at 20 °C, derived physical constant	0.01 … 0.05 ohm*mm^2/m
alpha	Temperature coefficient of resistance	0.00393	1/K	reference: record: annealed copper, 0.00393 per K at 20 °C, derived physical constant	0.003 … 0.0045 1/K
T_c	Conductor temperature for the resistance	70	degC	assumption: Cable warm from the running load before a restart.	20 … 90 degC

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
X_km	Reactance per km per conductor	0.08	ohm/km	reference: record: cable reactance 0.08 Ω/km (generic value as the E3 default; replace with the cable datasheet figure)	0 … 1 ohm/km
dip_limit	Dip limit at the motor terminals	10	%	spec: Machine specification MS-3: motor terminal voltage not below 90 % while starting	1 … 30 %
dip_limit_pcc	Dip limit at the point of common coupling (when the network operator sets one)	7	%	spec: Connection agreement CA-2: 7 % at the point of common coupling (test value invented for this example)	0.5 … 20 %
cos_phi2	Target power factor after correction	0.95		spec: Connection agreement CA-2: power factor not below 0.95	0.8 … 1
harmonics	Harmonic loads present on the supply	no		assumption: No drives or other harmonic loads on the supply.	any

3. Assumptions

A1. The supply is a stiff source behind the stated impedance (voltage E = U/√3 before the start). A direct-on-line or star-delta start, and a soft starter that does not reach its current limit at the dipped voltage (full conduction), draw current through a passive winding: a constant impedance equal to the locked-rotor impedance at rated voltage and the starting power factor, so the current falls with the dipped voltage. A soft starter that reaches its limit and a drive regulate the current: they hold |I| = I_start whatever the voltage, so the dip is deeper than the passive model gives; the current phasor lags the voltage at the starter or drive input by the stated power factor. (Standard steady-state starting-dip screening model; the regulated case is solved in closed form.)
A2. The dip is the steady-state phasor value at the instant of locked rotor; the current falls as the motor accelerates, so the dip is the worst case. No other load is running on the supply and no run-up dynamics, flicker (Pst) or tap-changer action is assessed. (Stated limitation.)
A3. A star-delta start draws one third of the direct-on-line current at rated voltage (and one third of the starting torque, which is not checked here); the current spike at the star-to-delta changeover is not modelled. (Star-delta connection ratio.)
A4. The cable resistance follows E3: from the resistivity, area and temperature, or from R′ per kilometre; X′ per kilometre is a user or record value. No cable table is embedded. (User or record values.)
A5. The capacitor kvar corrects the running power factor of the motor input power P/η at rated load; the capacitor sits at the point of common coupling, and the resonance order uses the short-circuit apparent power U²/|Z_s| there (|Z_s| stands in for X_s, so h reads 1 to 1.4 % low). Capacitor switching, detuning and self-excitation on disconnection are not assessed. (Standard power-factor-correction relation.)

A6. Source: E5 'Z_s' and 'I_f' are earth-fault-loop quantities (line to earth through the protective conductor), not the positive-sequence source impedance a motor start sees, so they are not linked; the fault level links E5 I_k3. E5 computes I_k3 with c = 1.1 and U_n = √3·U_0 (398.4 V at 230 V), so the I_k basis defaults c to 1.1 and U should be √3·U_0 for an exact match with E5's Z_T. E6 declares no impedance port, so the transformer rating and impedance are entered. (Linking decision.)
A7. The source impedance should be taken at the minimum fault level (the deepest dip); a transformer alone is an infinite bus behind it, which understates the dip by a few per cent and overstates the resonance order unless the upstream network fault level S_kQ is entered. (Stated limitation.)
A8. With a drive, the power-factor correction starts from the drive's displacement factor, not the motor's, the drive is itself a harmonic load, and the 'torque factor' and the voltage at the starter or drive input (not the motor terminals) are reported only for what they are. (Regulated-current starters.)
A9. XR_Q (Upstream network X/R ratio) is taken as 10. (Network X/R taken as 10, mostly reactive.)
A10. T_c (Conductor temperature for the resistance) is taken as 70 degC. (Cable warm from the running load before a restart.)
A11. harmonics (Harmonic loads present on the supply) is taken as no. (No drives or other harmonic loads on the supply.)

4. Working

```
[10] I_rated = P / (sqrt(3) * U * cos_phi * eta)
= 160 kW / (sqrt(3) × 400 V × 0.88 × 0.95)
= 276.24 A
```
Motor rated current from the rated output
```
[13] I_start = I_LR_r * I_rated / 3
= 7 × 276.24 A / 3
= 644.57 A
```
Starting current at rated voltage (star connection: one third of direct on line)
```
[14] S_start = sqrt(3) * U * I_start
= sqrt(3) × 400 V × 644.57 A
= 446.57 kVA
```
Starting apparent power
```
[20] Z_T = u_k * U^2 / S_n
= 4 % × (400 V)^2 / 500 kVA
= 0.0128 ohm
```
Transformer impedance magnitude
```
[21] R_T = Z_T / sqrt(1 + XR^2)
= 0.0128 ohm / sqrt(1 + 4.2^2)
= 0.0029647 ohm
```
Transformer resistance
```
[22] X_T = R_T * XR
= 0.0029647 ohm × 4.2
= 0.012452 ohm
```
Transformer reactance
```
[23] Z_Q = U^2 / S_kQ
= (400 V)^2 / 15 MVA
= 0.010667 ohm
```
Upstream network impedance magnitude (referred to the secondary)
```
[24] R_Q = Z_Q / sqrt(1 + XR_Q^2)
= 0.010667 ohm / sqrt(1 + 10^2)
= 0.0010614 ohm
```
Upstream network resistance
```
[25] X_Q = R_Q * XR_Q
= 0.0010614 ohm × 10
= 0.010614 ohm
```
Upstream network reactance

[26] R_src = R_T + R_Q
= 0.0029647 ohm + 0.0010614 ohm
= 0.0040261 ohm
Source resistance (transformer plus upstream network)
[27] X_src = X_T + X_Q
= 0.012452 ohm + 0.010614 ohm
= 0.023066 ohm
Source reactance (transformer plus upstream network)
[28] Z_src = hypot(R_src, X_src)
= hypot(0.0040261 ohm, 0.023066 ohm)
= 0.023414 ohm
Source impedance magnitude
[36] rho_T = rho_20 * (1 + alpha * (T_c - 20 degC))
= 0.01724 ohm·mm²/m × (1 + 0.00393 1/K × (70 degC - 20 degC))
= 0.020628 ohm·mm²/m
Resistivity at the operating temperature
[37] R_cab = rho_T * L_cab / (A_c * n_par)
= 0.020628 ohm·mm²/m × 60 m / (95 mm² × 1)
= 0.013028 ohm
Cable resistance (parallel paths combined)
[38] X_cab = X_km * L_cab / n_par
= 0.08 ohm/km × 60 m / 1
= 0.0048 ohm
Cable reactance
[39] Z_m = U / (sqrt(3) * I_start)
= 400 V / (sqrt(3) × 644.57 A)
= 0.35829 ohm
Motor starting impedance magnitude
[40] R_m = Z_m * cos_phi_st
= 0.35829 ohm × 0.3
= 0.10749 ohm
Motor starting resistance
[41] X_m = Z_m * sqrt(1 - cos_phi_st^2)
= 0.35829 ohm × sqrt(1 - 0.3^2)
= 0.34178 ohm
Motor starting reactance
[42] E_ph = E = U/√3 (reference phasor, angle 0°)
= 400 V / √3
= 230.94 V
Source phase voltage before the start
[43] Z_s_ph = Z_s = R_s + jX_s
= R_s = 0.0040261 □, X_s = 0.023066 □
= "0.0040261 + j0.023066 □ [0.023414 □ □ 80.099°]"
Source impedance phasor
[44] Z_c_ph = Z_c = R_c + jX_c
= R_c = 0.013028 □, X_c = 0.0048 □
= "0.013028 + j0.0048 □ [0.013884 □ □ 20.226°]"
Cable impedance phasor
[45] Z_m_ph = Z_m = |Z_m|·(cos □_st + j sin □_st)
= 0.35829 □ · (0.3 + j0.95394)
= "0.10749 + j0.34178 □ [0.35829 □ □ 72.542°]"
Motor starting impedance phasor

[46] Z_t_ph = Z_t = Z_s + Z_c + Z_m
= (0.0040261 + j0.023066) + (0.013028 + j0.0048) + (0.10749 + j0.34178)
= "0.12454 + j0.36965 □ [0.39006 □ □ 71.381°]"
Total series impedance phasor
[47] I_ph = I = E/Z_t
= 230.94 V / (0.39006 □ □ 71.381°)
= "189.03 - j561.07 A [592.06 A □ -71.381°]"
Starting current phasor at the dipped voltage
[48] V_m_ph = V_m = E·Z_m/Z_t
= 230.94 V · (0.35829 □ □ 72.542°) / (0.39006 □ □ 71.381°)
= "212.08 + j4.301 V [212.13 V □ 1.1618°]"
Motor terminal voltage phasor (phase)
[49] V_pcc_ph = V_pcc = E·(Z_c + Z_m)/Z_t
= 230.94 V · (0.36694 □ □ 70.826°) / (0.39006 □ □ 71.381°)
= "217.24 - j2.1012 V [217.25 V □ -0.55417°]"
Point of common coupling voltage phasor (phase)
[50] I_dip = |I|
= 230.94 V / 0.39006 □
= 592.06 A
Starting current at the dipped voltage
[51] V_m = |V_m|·√3
= 212.13 V · √3
= 367.41 V
Voltage at the motor terminals while starting (line)
[52] V_pcc = |V_pcc|·√3
= 217.25 V · √3
= 376.28 V
Point of common coupling voltage while starting (line)
[53] start_dip = 100·(1 - |V_m|/E)
= 100 · (1 - 212.13 / 230.94)
= 8.147 %
Voltage dip at the motor terminals
[54] dip_pcc = 100·(1 - |V_pcc|/E)
= 100 · (1 - 217.25 / 230.94)
= 5.9289 %
Voltage dip at the point of common coupling
[55] dV_cab = √3·Re[I·Z_c·V̄_m/|V_m|]
= √3 · Re[(592.06 A □ -71.381°) · (0.013884 □ □ 20.226°) · e^(-j1.1618°)]
= 8.7035 V
Cable drop along the starting voltage: √3·I·(R cos φ + X sin φ), the E3 three-phase form at this current
[56] k_T = (|V_m|/E)²
= (212.13 / 230.94)²
= 0.8437
Torque factor at the dipped voltage relative to rated voltage (torque varies as V²)
[57] u_dip = start_dip / dip_limit
= 8.147 % / 10 %
= 0.8147
Dip at the motor terminals against the limit
[58] P_in = P / eta
= 160 kW / 0.95
= 168.42 kW
Motor input power at rated load

[59] tan_1 = sqrt(1 - cos_phi^2) / cos_phi
= sqrt(1 - 0.88^2) / 0.88
= 0.53974
tan φ_1 at the running power factor
[60] tan_2 = sqrt(1 - cos_phi2^2) / cos_phi2
= sqrt(1 - 0.95^2) / 0.95
= 0.32868
tan φ_2 at the target power factor
[61] Q_c = max(0 kW, P_in * (tan_1 - tan_2))
= max(0 kW, 168.42 kW × (0.53974 - 0.32868))
= 35.547 kvar
Capacitor kvar to reach the target power factor
[62] I_cap = Q_c / (sqrt(3) * U)
= 35.547 kvar / (sqrt(3) × 400 V)
= 51.307 A
Capacitor line current at rated voltage
[63] I_run1 = P_in / (sqrt(3) * U * cos_phi)
= 168.42 kW / (sqrt(3) × 400 V × 0.88)
= 276.24 A
Running line current before correction
[64] I_run2 = P_in / (sqrt(3) * U * cos_phi2)
= 168.42 kW / (sqrt(3) × 400 V × 0.95)
= 255.89 A
Running line current after correction (when the target is reached)
[65] S_sc = U^2 / Z_src
= (400 V)^2 / 0.023414 ohm
= 6.8334 MVA
Short-circuit apparent power at the point of supply
[66] h_res = sqrt(S_sc / Q_c)
= sqrt(6.8334 MVA / 35.547 kvar)
= 13.865
Parallel resonance order of the capacitor bank with the supply
[68] u_dip_pcc = dip_pcc / dip_limit_pcc
= 5.9289 % / 7 %
= 0.84699
Point-of-common-coupling dip against the limit

5. Results

SYMBOL	VALUE	DESCRIPTION
Z_src	0.023414 ohm	Source impedance (magnitude)
Z_m	0.35829 ohm	Motor starting impedance (magnitude)
I_rated	276.24 A	Motor rated current
I_start	644.57 A	Starting current at rated voltage
S_start	446.57 kVA	Starting apparent power
I_dip	592.06 A	Starting current at the dipped voltage
V_m	367.41 V	Voltage at the motor terminals while starting
V_pcc	376.28 V	Point of common coupling voltage while starting
start_dip	8.147 %	Voltage dip at the motor terminals
dip_pcc	5.9289 %	Voltage dip at the point of common coupling

SYMBOL	VALUE	DESCRIPTION
dV_cab	8.7035 V	Cable drop along the starting voltage
k_T	0.8437	Torque factor at the dipped voltage
u_dip	0.8147	Dip utilisation at the motor terminals
P_in	168.42 kW	Motor input power at rated load
kvar	35.547 kvar	Capacitor kvar to the target power factor
I_cap	51.307 A	Capacitor current
I_run1	276.24 A	Running current before correction
I_run2	255.89 A	Running current after correction
S_sc	6.8334 MVA	Short-circuit apparent power at the bank
h_res	13.865	Parallel resonance order
u_dip_pcc	0.84699	Point-of-common-coupling dip utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Point-of-common-coupling dip within the limit	u_dip_pcc = 0.84699	u_dip_pcc ≤ 1	0.847	IEC 61000-3-11 scope
PASS	Motor-terminal dip within the limit	u_dip = 0.8147	u_dip ≤ 1	0.815	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: X_km (Cable resistance and reactance per kilometre); rho_20, alpha (Conductor material constants); I_LR_r, cos_phi_st (Motor starting data); u_k, XR (Transformer impedance data). Enter each as one so its source, edition and licence travel with it.
WARNING — input.fault_basis: fault_basis (Fault level given as) was not entered; the template default 'Fault level S_k (MVA)' is used.
WARNING — input.S_k: S_k (Three-phase fault level at the point of supply (use the minimum fault level, which gives the deepest dip)) was not entered; the template default 10 MVA is used.
WARNING — input.I_k: I_k (Three-phase fault current at the point of supply (links E5 I_k3)) was not entered; the template default 15 kA is used.
WARNING — input.k_soft: k_soft (Soft-starter current limit as a multiple of the rated current) was not entered; the template default 3.5 is used.
WARNING — input.k_vfd: k_vfd (Drive line-current limit as a multiple of the rated current) was not entered; the template default 1.5 is used.
WARNING — input.cos_phi_vfd: cos_phi_vfd (Drive line displacement power factor while starting) was not entered; the template default 0.95 is used.

8. Verification

Template elec.motor-start@1.2.0 Content hash 619b634f509eb509 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0